

FITBUDDY AI

Phase 1: Brainstorming & Ideation

Prepared from the current public FitBuddy_ai repository and organized according to the Phase 1–6 structure of the provided AI/ML & GenAI project template.

1.1 Project Overview

FitBuddy AI is a database-backed fitness assistant web application. The public repository describes it as an application intended to generate personalized workout programs and support nutritional guidance based on user biometrics, while providing an interactive feedback pipeline for plan adjustment.

1.2 Problem Statement

People often need a simple way to organize workout activity around their age, body measurements, fitness goal, experience level, and available workout days. A conventional static plan does not provide a convenient feedback loop for modifying the plan after the user tries it.

1.3 Proposed Solution

FitBuddy provides a web interface where a user registers a fitness profile, receives a generated workout blueprint, views a dashboard containing the profile and plan, submits feedback, and receives an updated plan record. The application also provides an administrative view of stored user plans.

1.4 Objectives

- Collect structured fitness-profile information through a browser form.
- Generate a personalized workout blueprint from the submitted profile.
- Persist the original plan and user profile in SQLite.
- Allow the user to submit feedback about the plan.
- Store an adapted plan together with the feedback.
- Provide a user dashboard and an all-users administrative registry.
- Provide a foundation for future Gemini-based generative AI integration.

1.5 Target Users

- General users who want a structured fitness-plan starting point.
- Users who want to provide feedback and keep an updated plan.
- Project administrators who need to inspect stored profile and plan records.

1.6 Key Features

Feature	FitBuddy implementation / intent
Profile registration	Collects name, age, gender, weight, height, workout days, goal and experience level.
Workout blueprint	Current public route generates a deterministic workout schedule from the submitted profile.

Feature	FitBuddy implementation / intent
Dashboard	Displays profile information, nutrition guidance, current plan and feedback form.
Feedback loop	Stores feedback and creates an adapted-plan record based on the original plan.
Admin registry	Lists users, metrics, original plans, updated plans and feedback.
SQLite storage	Stores the user profile and plan-related fields in a users table.
GenAI foundation	Configuration includes Gemini model settings; direct Gemini invocation is a future integration item in the visible route implementation.

1.7 Initial Workflow

User profile → FastAPI route → workout blueprint → SQLite → dashboard → feedback → updated plan → admin registry

1.8 Innovation / Project Value

- Combines profile-based fitness planning with persistent records.
- Includes a feedback-driven update mechanism rather than a one-time output.
- Provides both user-facing and administrative views.
- Can be extended from deterministic plan generation to Gemini-powered personalized generation.

Project repository: https://github.com/asvikamoorthy-cyber/FitBuddy_ai

Reference structure: <https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template>

Scope note: This document describes the project state visible in the public repository and identifies proposed/future items separately where applicable.